



Exploring Trends in the Automotive Industry Using SQL

A data analytics portfolio project analyzing over 8,000 used-car records to uncover market trends, pricing patterns, and business insights.

MYSQL

MYSQL WORKBENCH

SQL

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Project Overview

Analyze more than **8,000 used-car records** using SQL to identify market trends and pricing patterns across fuel types, ownership, and mileage.



Analyze Vehicle Prices

Examine selling price distributions across brands and model years.



Compare Fuel Types

Contrast pricing and performance across petrol, diesel, CNG, LPG, and electric vehicles.



Study Ownership Impact

Measure how first, second, and third ownership affects resale value.

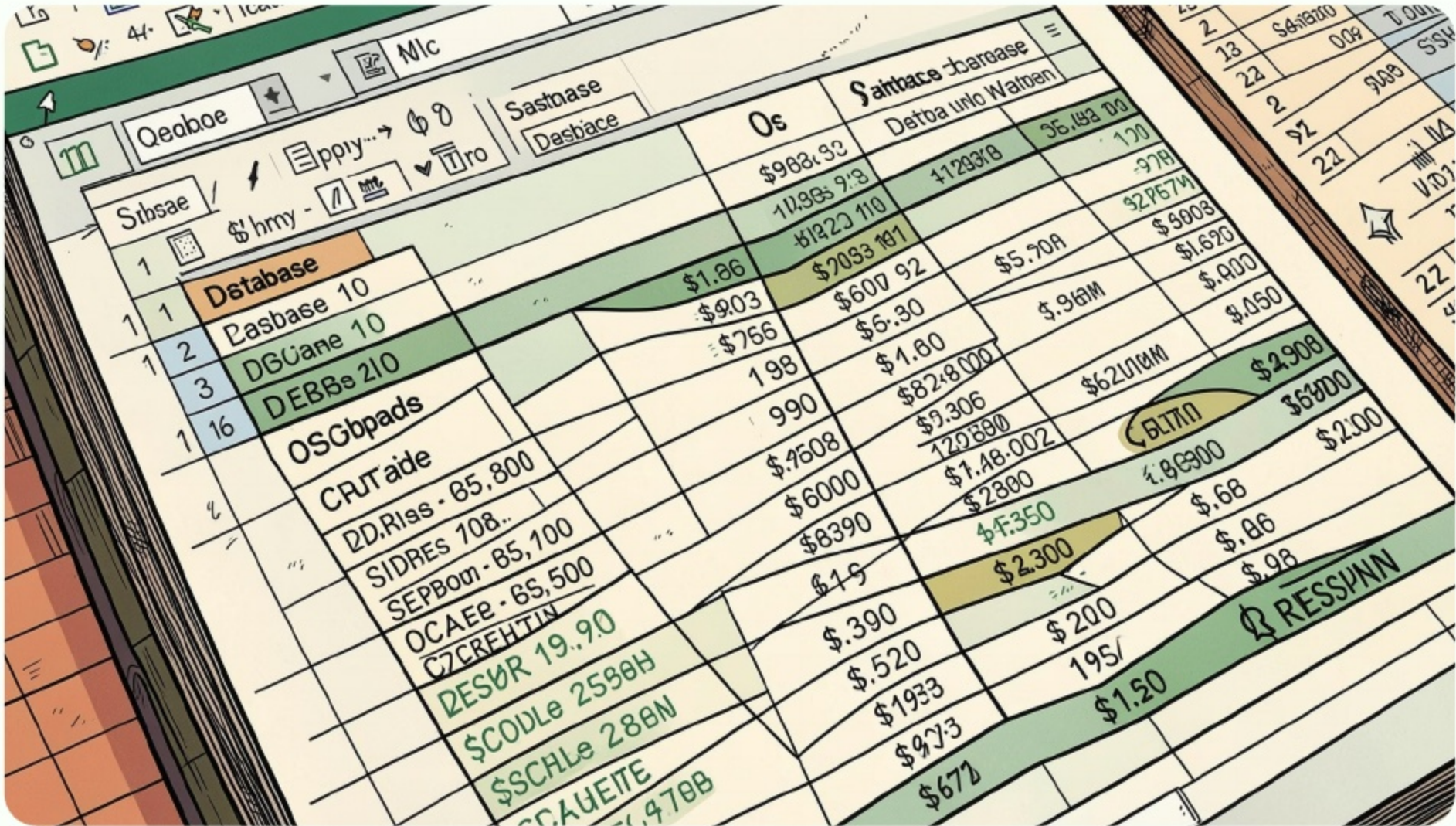


Understand Mileage Trends

Explore the relationship between mileage, engine capacity, and vehicle pricing.



Dataset Information



Dataset Summary

Total Records	8,148
Columns	13
Fuel Types	Petrol, Diesel, CNG, LPG, Electric
Transmission	Manual & Automatic
Ownership	First, Second, Third Owner

Key Columns Analyzed

Name · Year · Selling Price · Fuel · Mileage · Engine · Transmission · Seats · Owner

Tools & SQL Concepts Used

Tools

MySQL

Relational database engine for querying and managing the dataset.

MySQL Workbench

Visual IDE for writing, testing, and visualizing SQL queries.

SQL Concepts Applied

- **SELECT, WHERE, ORDER BY** – Filtering and sorting records
- **GROUP BY & HAVING** – Aggregating and filtering grouped results
- **Aggregate Functions** – COUNT, SUM, AVG, MIN, MAX
- **CASE & Subqueries** – Conditional logic and nested queries
- **CTEs & Window Functions** – Advanced analytical queries

Project Workflow

A structured six-stage pipeline transformed raw automotive data into actionable business intelligence using SQL at every step.



Each stage built upon the previous, ensuring data quality before drawing conclusions and recommendations.

Basic Analysis

8,148

Total Cars

Records in the dataset

5

Fuel Types

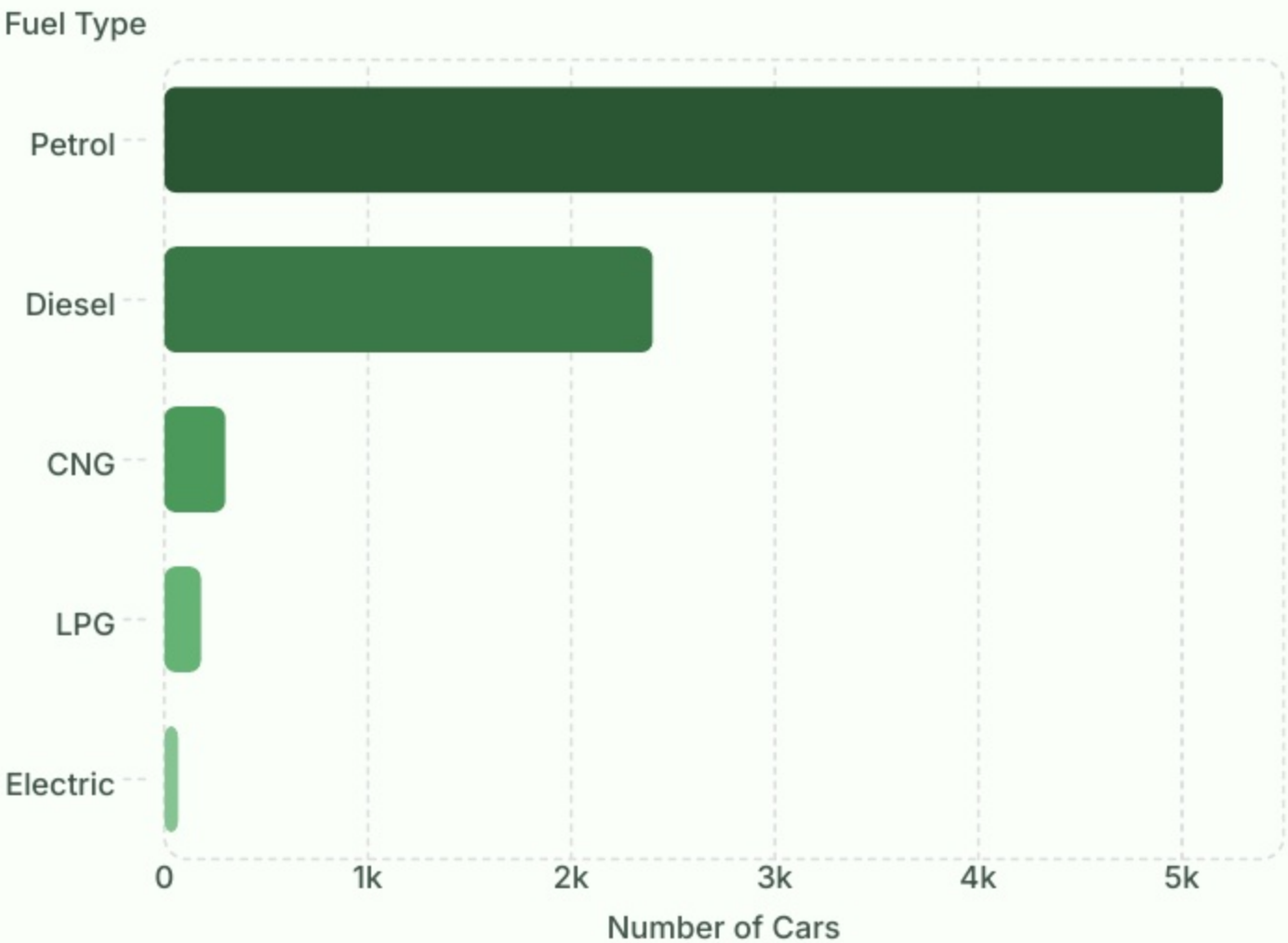
Petrol, Diesel, CNG, LPG, Electric

2

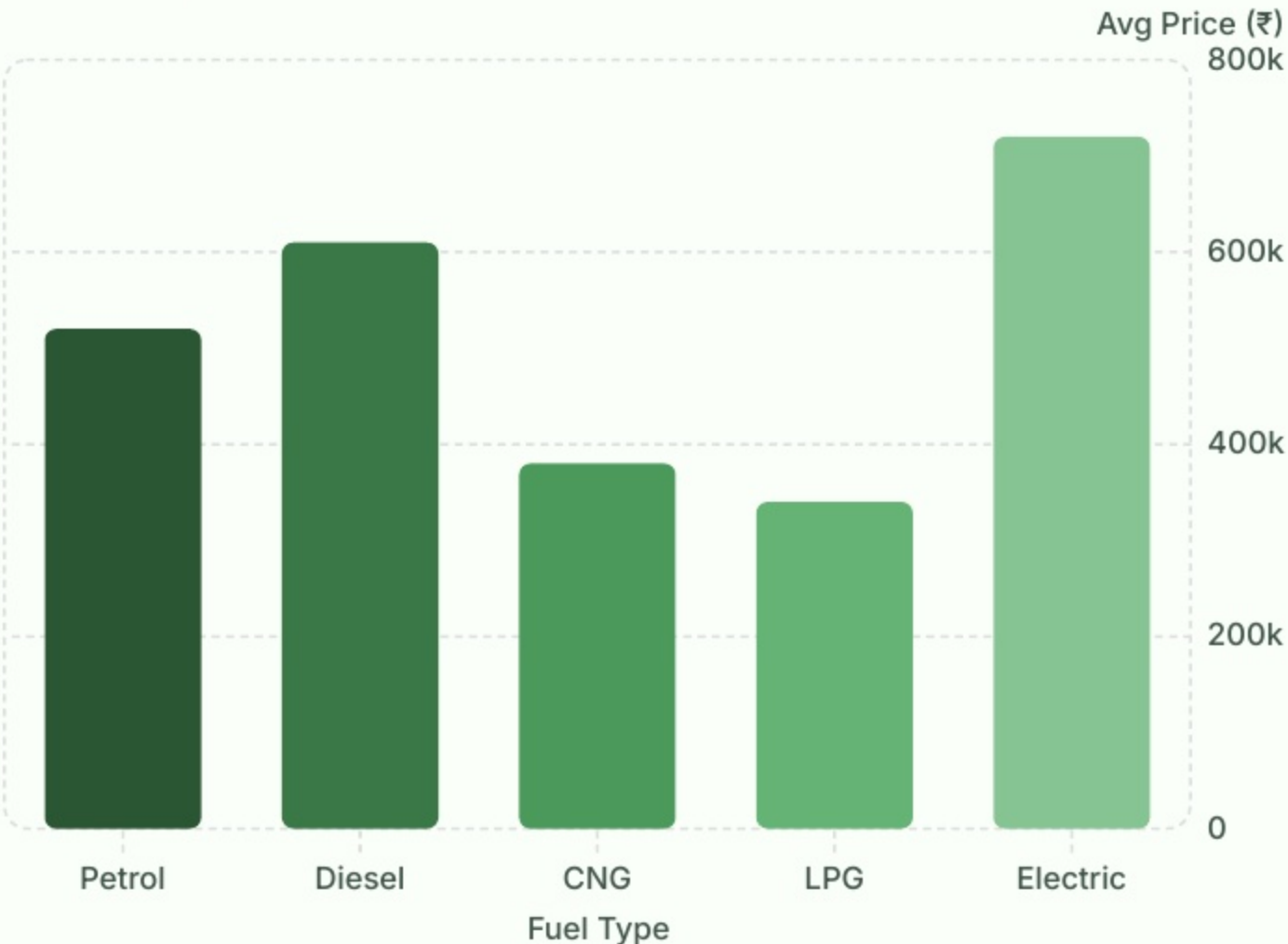
Transmission Types

Manual & Automatic

Cars by Fuel Type

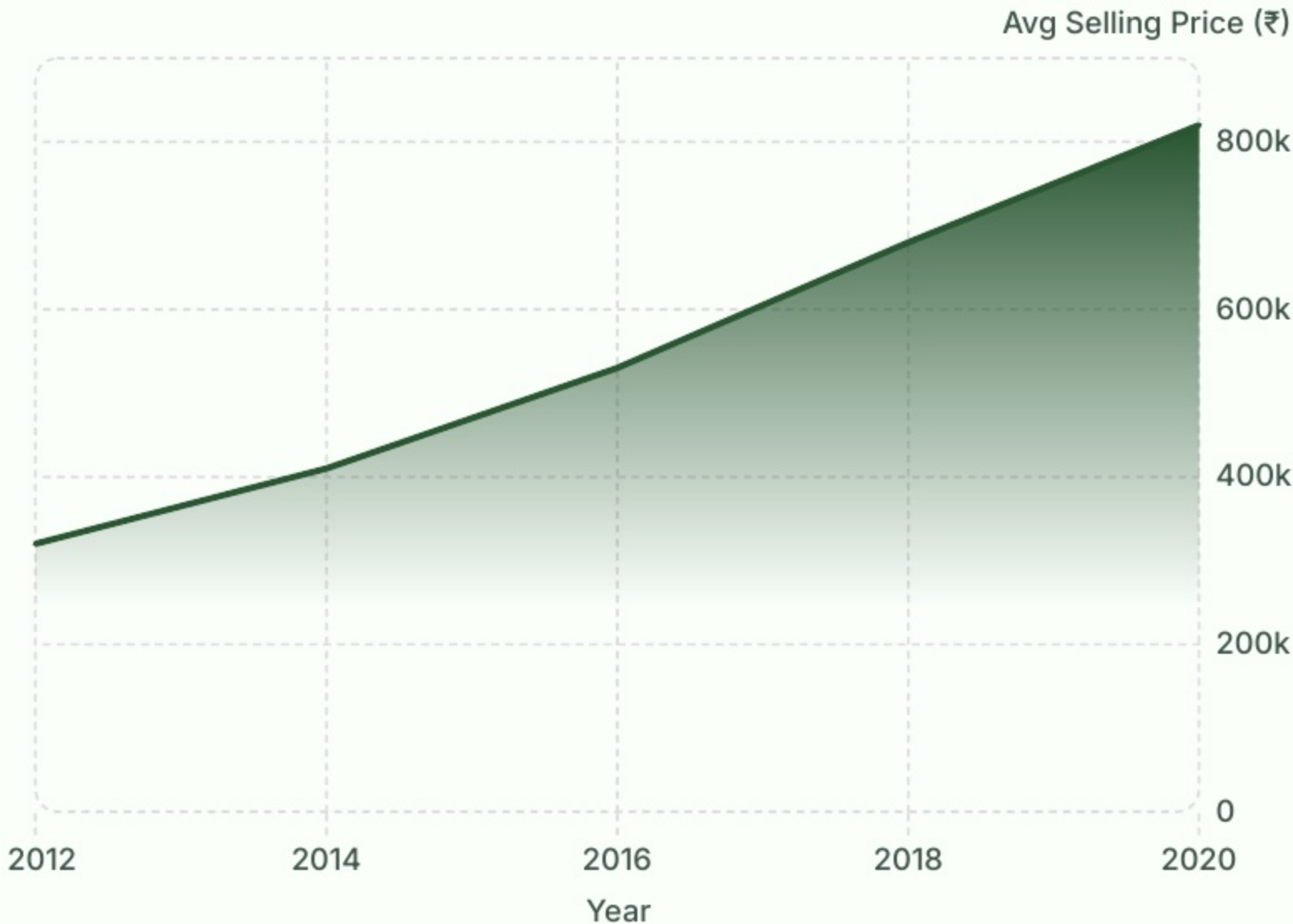


Average Price by Fuel Type (₹)



Vehicle Price Analysis

Selling Price Trend by Year



Top 5 Highest Average Selling Price Models

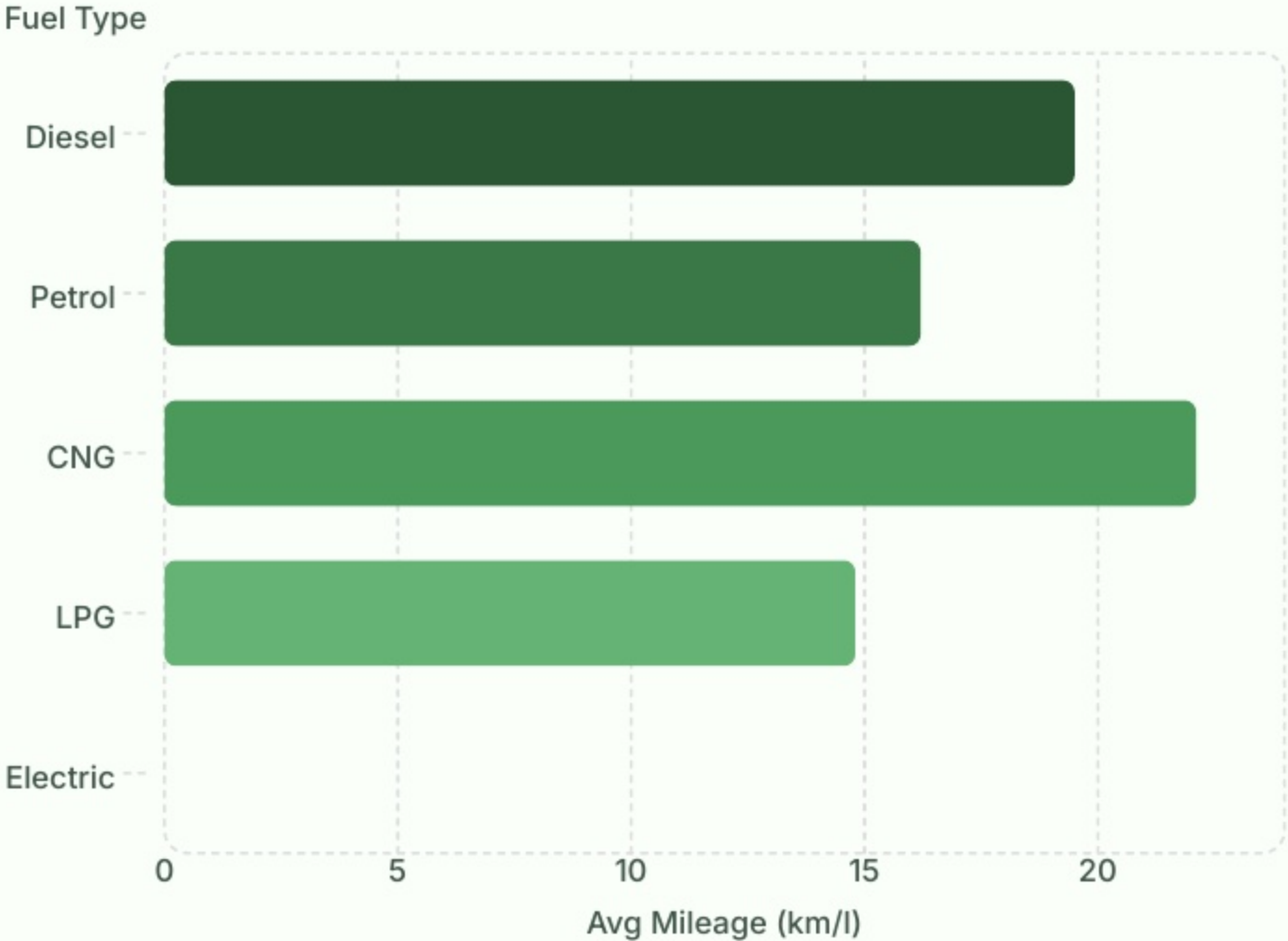
Model	Avg Price (₹)
Mercedes-Benz C-Class	1,850,000
BMW 3 Series	1,620,000
Audi A4	1,480,000
Toyota Fortuner	1,320,000
Honda CR-V	1,150,000



Luxury and SUV segments command the highest average resale values, reflecting brand prestige and market demand.

Mileage & Performance

Average Mileage by Fuel Type (km/l)



Key Insights

 Diesel Leads

Diesel vehicles generally provide better mileage among conventional fuel types, making them a preferred choice for high-usage buyers.

 CNG Efficiency

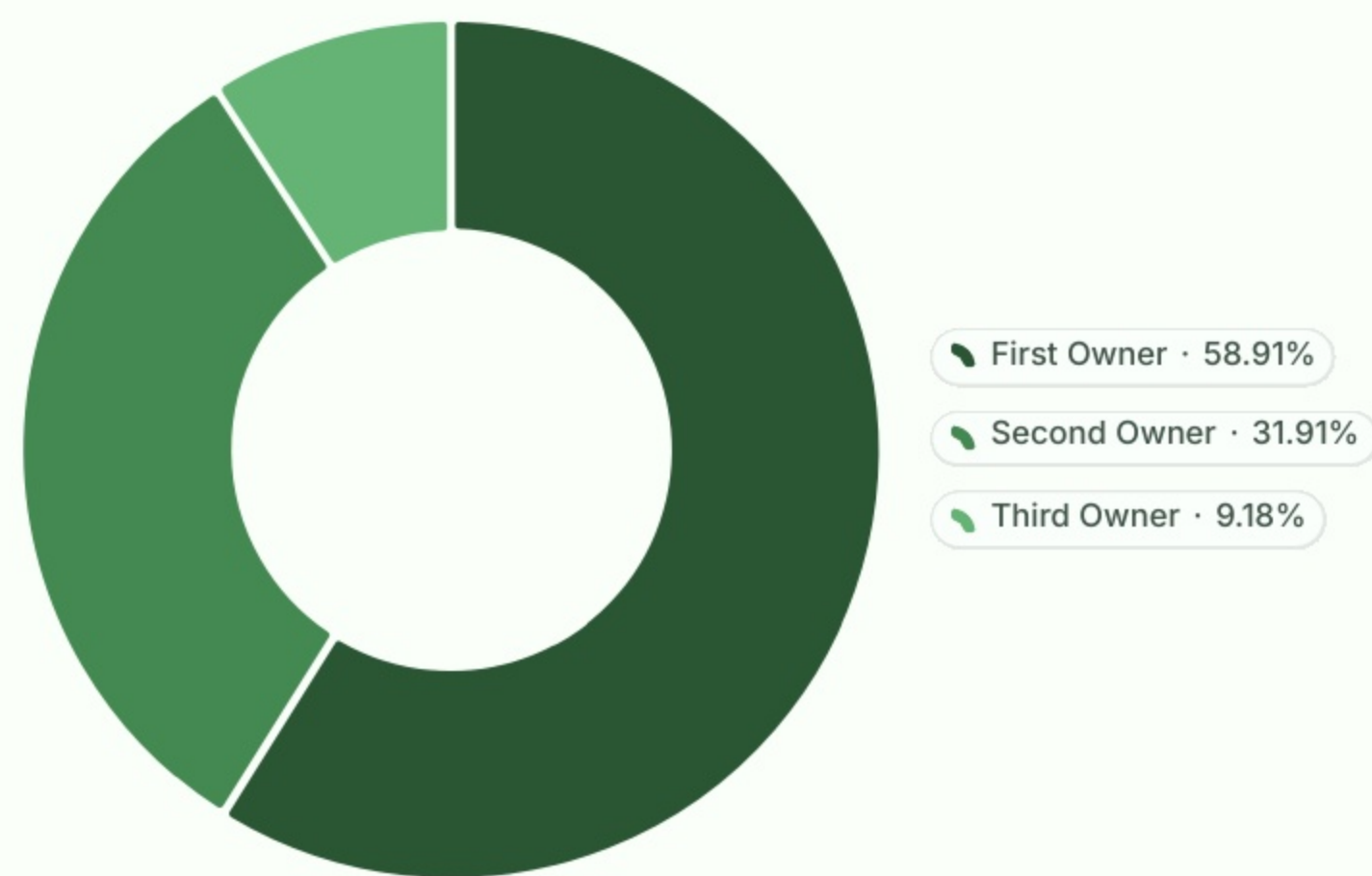
CNG vehicles show the highest km/l figures, though limited model availability affects overall market share.

 Engine Capacity

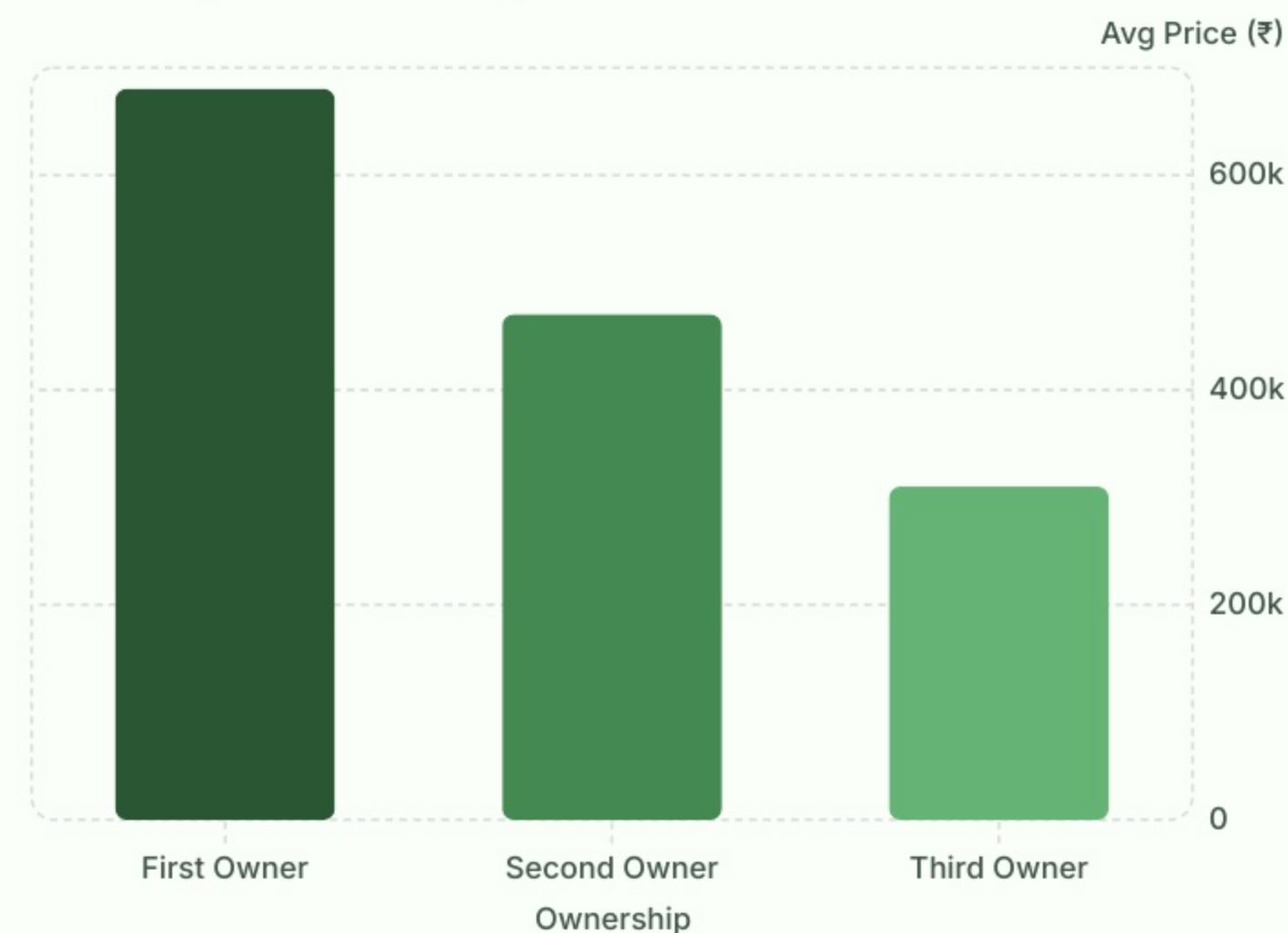
Larger engine capacities correlate with lower mileage but higher average selling prices, particularly in the SUV segment.

Ownership Analysis

Ownership Distribution



Average Price by Owner (₹)



Key Finding: First-owner vehicles command the highest resale value – approximately **45% more** than third-owner cars, reflecting better maintenance history and lower perceived risk.

Advanced SQL Techniques

Techniques Demonstrated

Subqueries

CTEs

LAG()

RANK()

SUM() OVER()

AVG() OVER()

Example: LAG() for Depreciation Analysis

This query compares each year's average price to the previous year, revealing depreciation trends.

```
WITH yearly_avg AS (  
  SELECT Year,  
         AVG(Selling_Price) AS avg_price  
  FROM cars  
  GROUP BY Year  
)  
SELECT Year,  
       avg_price,  
       LAG(avg_price) OVER (  
         ORDER BY Year) AS prev_year_price  
FROM yearly_avg;
```


Key Business Insights

These actionable insights, derived from our comprehensive SQL analysis, highlight critical market dynamics in the used-car industry.



Newer Vehicles, Higher Prices

More recent model years consistently command higher average selling prices, indicating a strong preference for newer cars.



Diesel's Mileage Advantage

Diesel cars demonstrate superior fuel efficiency (km/l) compared to petrol variants, appealing to buyers prioritising economy.



Automatic Commands More

Automatic transmission vehicles are priced higher on average than manual counterparts, reflecting convenience and modern preference.



Luxury Market Premium

Premium brands such as BMW dominate the highest price segments, reflecting their strong brand value and perceived quality.



First-Owner Value Retention

Vehicles with a single previous owner retain significantly higher resale value, suggesting trust in maintenance history.



Age Affects Depreciation

Vehicle age is a critical factor in depreciation, with prices declining consistently as cars get older.

Conclusion & Thank You

Our comprehensive SQL analysis transformed raw data into critical market intelligence, offering a clear understanding of the used-car industry dynamics.

1

Data Mastery

Analyzed over 8,000 used-car records for a robust dataset foundation.

2

SQL Prowess

Developed 30+ advanced SQL queries utilizing CTEs, window functions, and subqueries.

3

Market Insights

Identified key trends in pricing, mileage, ownership, and fuel type performance.

4

Actionable Intelligence

Generated business insights crucial for strategic decision-making in the automotive sector.

Thank You